

1. Encapsulation means keeping data private and controlling access through methods. Private variables, like radius, are encapsulated.
2. The constructor must have the same exact name as the class.
Example:

```
public class Circle {  
    public Circle() {  
        // constructor  
    }  
}
```
3. Private access modifiers can only be accessed inside the same class and are used to protect data. Public access modifiers can be accessed from anywhere, including other classes.
4. Invalid because radius is usually declared private. You can't access private variables from outside the class.
5.
 - a) Roo.
 - b) x
 - c) Accessor = gets value
 getX()
 - d) Modifier = changes value
 setX(int z)
 - e) Helper = used internally
 factor()
 - f) Roo
 - g) 5 methods.
6. A class is a blueprint, and defines variables and methods. An object is a real instance of a class.
9.
 - a) z because static final means it is a constant.
 - b) y and x.
 - c) y is a data member and an instance member.
 - d) x and z are class members.
11. Overloading used the same method name, but different inputs. It happens within the same class and is used to make methods more flexible. Overriding uses the same method name and same inputs, it happens in different classes and the child class replaces the parent class method.